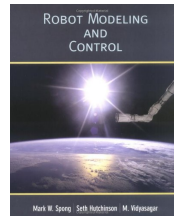


Quaternions

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Quaternions

A quaternion is a four-dimensional complex number defined as the sum of a scalar q_0 and a vector $\mathbf{q} = (q_1, q_2, q_3)$:

$$\mathbf{Q} = q_0 + \mathbf{q} = q_0 + iq_1 + jq_2 + kq_3 \quad (1)$$

A quaternion has the following properties:

$$ii = jj = kk = ijk = -1$$

$$ij = -ji = k$$

$$jk = -kj = i$$

$$ki = -ik = j$$

Quaternions

The multiplication of two quaternions is defined as follows:

$$\begin{aligned}\mathbf{Q} \otimes \mathbf{P} &= (q_0 + iq_1 + jq_2 + kq_3)(p_0 + ip_1 + jp_2 + kp_3) \\ &= (p_0q_0 - \mathbf{p} \cdot \mathbf{q}, p_0\mathbf{q} + q_0\mathbf{p} + \mathbf{q} \times \mathbf{p})\end{aligned}\tag{2}$$

where $\mathbf{q} = (q_1, q_2, q_3)$,

$\mathbf{p} = (p_1, p_2, p_3)$,

$\mathbf{p} \cdot \mathbf{q}$ = dot product of $\mathbf{p}$ and $\mathbf{q}$, and

$\mathbf{q} \times \mathbf{p}$ = cross product of $\mathbf{q}$ and $\mathbf{p}$ ¹.

¹see Appendix.

Quaternions

Example. Find the result of the following multiplication of quaternions:

$$\mathbf{Q} \otimes \mathbf{P} = (2 - i + j + 3k)(-1 + i + 4j - 2k)$$

Quaternions

Solution.

$$\mathbf{Q} = (2 - i + j + 3k)$$

$$\mathbf{P} = (-1 + i + 4j - 2k)$$

$$p_0q_0 = (-1)(2) = -2$$

$$\mathbf{p} \cdot \mathbf{q} = (1, 4, -2) \cdot (-1, 1, 3) = -1 + 4 - 6 = -3$$

$$p_0\mathbf{q} = (-1)(-i + j + 3k) = i - j - 3k$$

$$q_0\mathbf{p} = (2)(i + 4j - 2k) = 2i + 8j - 4k$$

$$\mathbf{q} \times \mathbf{p} = (-2 - 12)i + (3 - 2)j + (-4 - 1)k = -14i + j - 5k$$

$$\begin{aligned}\mathbf{Q} \otimes \mathbf{P} &= -2 - (-3), (i - j - 3k) + (2i + 8j - 4k) + (-14i + j - 5k) \\ &= 1 - 11i + 8j - 12k\end{aligned}$$

Quaternions

Example. Find the result of the following multiplication of quaternions:

$$\mathbf{Q} \otimes \mathbf{P} = (-2 + 3i - 5j + k)(2 + i - 4j + 2k)$$

Quaternions

Solution.

$$\mathbf{Q} = (-2 + 3i - 5j + k)$$

$$\mathbf{P} = (2 + i - 4j + 2k)$$

$$p_0q_0 = (-2)(2) = -4$$

$$\mathbf{p} \cdot \mathbf{q} = (1, -4, 2) \cdot (3, -5, 1) = 3 + 20 + 2 = 25$$

$$p_0\mathbf{q} = (2)(3i - 5j + k) = 6i - 10j + 2k$$

$$q_0\mathbf{p} = (-2)(i - 4j + 2k) = -2i + 8j - 4k$$

$$\mathbf{q} \times \mathbf{p} = -6i - 5j - 7k$$

$$\begin{aligned}\mathbf{Q} \otimes \mathbf{P} &= -4 - 25, (6i - 10j + 2k) + (-2i + 8j - 4k) + (-6i - 5j - 7k) \\ &= -29 - 2i - 7j - 9k\end{aligned}$$

Quaternions

The complex conjugate of a quaternion is defined as:

$$\mathbf{Q}^* = (q_0 + q_1i + q_2j + q_3k)^* = (q_0 - q_1i - q_2j - q_3k)$$

Quaternions

The complex conjugate of a quaternion is defined as:

$$\mathbf{Q} = (q_0 + q_1i + q_2j + q_3k)^* = (q_0 - q_1i - q_2j - q_3k)$$

Exercise. Find the complex conjugate of the following quaternions:

$$\mathbf{Q} = (5 + 2i + 3j - k)$$

$$\mathbf{Q} = (-3 - 3i + 2j + 5k)$$

$$\mathbf{Q} = (2 - 2i - 0.5j - 7k)$$

Quaternions

The complex conjugate of a quaternion is defined as:

$$\mathbf{Q} = (q_0 + q_1i + q_2j + q_3k)^* = (q_0 - q_1i - q_2j - q_3k)$$

Solution.

$$\mathbf{Q}^* = (5 - 2i - 3j + k)$$

$$\mathbf{Q}^* = (-3 + 3i - 2j - 5k)$$

$$\mathbf{Q}^* = (2 + 2i + 0.5j + 7k)$$

Quaternions

The unit quaternion is defined as follows:

$$\text{norm}(\hat{\mathbf{Q}}) = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} = 1 \quad (3)$$

Quaternions

The unit quaternion is defined as follows:

$$\text{norm}(\hat{\mathbf{Q}}) = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} = 1 \quad (4)$$

Exercise. Are they unit quaternions?

$$\hat{\mathbf{Q}} = (1 - 2i - 5j + 3k)$$

$$\hat{\mathbf{Q}} = (0.1 - 0.2i - 0.3j + 0.4k)$$

$$\hat{\mathbf{Q}} = \left(\frac{3}{\sqrt{23}} - \frac{2}{\sqrt{23}}i - \frac{3}{\sqrt{23}}j + \frac{1}{\sqrt{23}}k \right)$$

$$\hat{\mathbf{Q}} = \left(\frac{4}{\sqrt{30}} - \frac{2}{\sqrt{30}}i - \frac{3}{\sqrt{30}}j + \frac{1}{\sqrt{30}}k \right)$$

Quaternions

The unit quaternion is defined as follows:

$$\text{norm}(\hat{\mathbf{Q}}) = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} = 1 \quad (5)$$

Exercise. Are they unit quaternions?

$$\hat{\mathbf{Q}} = (1 - 2i - 5j + 3k)$$

No

$$\hat{\mathbf{Q}} = (0.1 - 0.2i - 0.3j + 0.4k)$$

No

$$\hat{\mathbf{Q}} = \left(\frac{3}{\sqrt{23}} - \frac{2}{\sqrt{23}}i - \frac{3}{\sqrt{23}}j + \frac{1}{\sqrt{23}}k \right)$$

Yes

$$\hat{\mathbf{Q}} = \left(\frac{4}{\sqrt{30}} - \frac{2}{\sqrt{30}}i - \frac{3}{\sqrt{30}}j + \frac{1}{\sqrt{30}}k \right)$$

Yes

Quaternions

The inverse of a unit quaternion is equal to its complex conjugate:

$$\hat{\mathbf{Q}}^{-1} = \hat{\mathbf{Q}}^* = (q_0 - q_1 - q_2 - q_3) \quad (6)$$

The inverse of a non-unit quaternion is equal to

$$\mathbf{Q}^{-1} = \frac{\mathbf{Q}^*}{||\mathbf{Q}||^2} \quad (7)$$

Quaternions

Any unit quaternion $\hat{\mathbf{Q}}$ may be written as

$$\hat{\mathbf{Q}} = q_0 + \mathbf{q} = \cos\left(\frac{\theta}{2}\right) + \mathbf{u}\sin\left(\frac{\theta}{2}\right) \quad (8)$$

where

$$\mathbf{u} = \frac{\mathbf{q}}{\|\mathbf{q}\|} \quad \text{and} \quad \tan\left(\frac{\theta}{2}\right) = \frac{\|\mathbf{q}\|}{q_0}.$$

Quaternions

Explanation 1/1.

A unit quaternion must satisfy that $q_0^2 + \|\mathbf{q}\|^2 = 1$. So, there must exist some angle θ such that

$$\cos^2\left(\frac{\theta}{2}\right) = q_0^2 \quad \text{and} \quad \sin^2\left(\frac{\theta}{2}\right) = \|\mathbf{q}\|^2$$

It also implies that $\cos\left(\frac{\theta}{2}\right) = q_0$ and $\sin\left(\frac{\theta}{2}\right) = \|\mathbf{q}\|$.

Therefore, the unit quaternion $\hat{\mathbf{Q}} = q_0 + \mathbf{q}$ can now be written in terms of the angle $\left(\frac{\theta}{2}\right)$ and the unit vector $\mathbf{u}$:

$$\hat{\mathbf{Q}} = q_0 + \mathbf{q} = \cos\left(\frac{\theta}{2}\right) + \mathbf{u}\sin\left(\frac{\theta}{2}\right)$$

Quaternions

We can use a unit quaternion $\hat{\mathbf{Q}} = q_0 + \mathbf{q}$ to define a linear operator $L_{\hat{\mathbf{Q}}}(\mathbf{v})$ that acts on vector $\mathbf{v}$ as follows:

$$\begin{aligned} L_{\hat{\mathbf{Q}}}(\mathbf{v}) &= \hat{\mathbf{Q}}\mathbf{v}\hat{\mathbf{Q}}^* \\ &= (q_0^2 - \|\mathbf{q}\|^2) \mathbf{v} + 2(\mathbf{q} \cdot \mathbf{v})\mathbf{q} + 2q_0 (\mathbf{q} \times \mathbf{v}) \end{aligned} \quad (9)$$

Quaternions

We can use a unit quaternion $\hat{\mathbf{Q}} = q_0 + \mathbf{q}$ to define a linear operator $L_{\hat{\mathbf{Q}}}(\mathbf{v})$ that acts on vector $\mathbf{v}$ as follows:

$$\begin{aligned} L_{\hat{\mathbf{Q}}}(\mathbf{v}) &= \hat{\mathbf{Q}}\mathbf{v}\hat{\mathbf{Q}}^* \\ &= (q_0^2 - \|\mathbf{q}\|^2) \mathbf{v} + 2(\mathbf{q} \cdot \mathbf{v})\mathbf{q} + 2q_0 (\mathbf{q} \times \mathbf{v}) \end{aligned} \quad (34)$$

Homework. Prove equation (9).

Quaternions

Proof 1/3. Consider $\mathbf{v} = 0 + \mathbf{v}$, $\hat{\mathbf{Q}} = q_0 + \mathbf{q}$ and $\hat{\mathbf{Q}}^* = q_0 - \mathbf{q}$. Therefore, we have that

$$\begin{aligned} L_{\hat{\mathbf{Q}}}(\mathbf{v}) &= \hat{\mathbf{Q}}\mathbf{v}\hat{\mathbf{Q}}^* \\ &= (q_0 + \mathbf{q})\mathbf{v}(q_0 - \mathbf{q}) \\ &= ((q_0(0) - \mathbf{q} \cdot \mathbf{v}) + (q_0\mathbf{v} + \mathbf{q} \times \mathbf{v}))(q_0 - \mathbf{q}) \\ &= (-\mathbf{q} \cdot \mathbf{v} + (q_0\mathbf{v} + \mathbf{q} \times \mathbf{v}))(q_0 - \mathbf{q}) \end{aligned}$$

From the equation above we see that $\hat{\mathbf{Q}}\mathbf{v}$ gives rise to a new quaternion $\mathbf{R} = r_0 + \mathbf{r} = -\mathbf{q} \cdot \mathbf{v} + (q_0\mathbf{v} + \mathbf{q} \times \mathbf{v})$.

$$r_0 = -\mathbf{q} \cdot \mathbf{v}$$

$$\mathbf{r} = q_0\mathbf{v} + \mathbf{q} \times \mathbf{v}$$

Quaternions

If we now analyze the product $(\hat{\mathbf{Q}}\mathbf{v}) \hat{\mathbf{Q}}^*$ we see that

$$\begin{aligned}(\hat{\mathbf{Q}}\mathbf{v}) \hat{\mathbf{Q}}^* &= r_0 q_0 - (\mathbf{r} \cdot (-\mathbf{q})) + r_0(-\mathbf{q}) + q_0 \mathbf{r} + \mathbf{r} \times (-\mathbf{q}) \\&= (\mathbf{q} \cdot \mathbf{v})\mathbf{q} + q_0^2 \mathbf{v} + q_0(\mathbf{q} \times \mathbf{v}) + q_0(\mathbf{q} \times \mathbf{v}) + (\mathbf{q} \cdot \mathbf{v})\mathbf{q} - \|\mathbf{q}\|^2 \mathbf{v} \\&= (q_0^2 - \|\mathbf{q}\|^2) \mathbf{v} + 2(\mathbf{q} \cdot \mathbf{v})\mathbf{q} + 2q_0 (\mathbf{q} \times \mathbf{v})\end{aligned}$$

Consider that

$$\begin{aligned}r_0 q_0 - (\mathbf{r} \cdot (-\mathbf{q})) &= r_0 q_0 + (\mathbf{r} \cdot \mathbf{q}) \\&= r_0 q_0 + (q_0 \mathbf{v} + \mathbf{q} \times \mathbf{v}) \cdot \mathbf{q} \\&= r_0 q_0 + (q_0 \mathbf{v}) \cdot \mathbf{q} + (\mathbf{q} \times \mathbf{v}) \cdot \mathbf{q} \\&= r_0 q_0 + q_0(\mathbf{v} \cdot \mathbf{q}) + (\mathbf{q} \times \mathbf{v}) \cdot \mathbf{q} \\&= -(\mathbf{q} \cdot \mathbf{v})q_0 + q_0(\mathbf{v} \cdot \mathbf{q}) + (\mathbf{q} \times \mathbf{v}) \cdot \mathbf{q} \\&= -q_0(\mathbf{q} \cdot \mathbf{v}) + q_0(\mathbf{q} \cdot \mathbf{v}) + (\mathbf{q} \times \mathbf{v}) \cdot \mathbf{q} = 0\end{aligned}$$

Quaternions

Additionally consider that:

$$\begin{aligned}r_0(-\mathbf{q}) &= (-\mathbf{q} \cdot \mathbf{v})(-\mathbf{q}) \\ &= (\mathbf{q} \cdot \mathbf{v})\mathbf{q}\end{aligned}$$

$$\begin{aligned}q_0\mathbf{r} &= q_0(q_0\mathbf{v}) + q_0(\mathbf{q} \times \mathbf{v}) \\ &= q_0^2\mathbf{v} + q_0(\mathbf{q} \times \mathbf{v})\end{aligned}$$

$$\begin{aligned}\mathbf{r} \times (-\mathbf{q}) &= (q_0\mathbf{v} + \mathbf{q} \times \mathbf{v}) \times (-\mathbf{q}) \\ &= (q_0\mathbf{v}) \times (-\mathbf{q}) + (\mathbf{q} \times \mathbf{v}) \times (-\mathbf{q}) \\ &= q_0(\mathbf{q} \times \mathbf{v}) + \mathbf{q} \times (\mathbf{q} \times \mathbf{v}) \\ &= q_0(\mathbf{q} \times \mathbf{v}) + (\mathbf{q} \cdot \mathbf{v})\mathbf{q} - (\mathbf{q} \cdot \mathbf{q})\mathbf{v} \\ &= q_0(\mathbf{q} \times \mathbf{v}) + (\mathbf{q} \cdot \mathbf{v})\mathbf{q} - \|\mathbf{q}\|^2\mathbf{v}\end{aligned}$$

Q.E.D.

Quaternions

If $\mathbf{v}$ is along $\mathbf{q}$ (i.e., $\mathbf{v} = k\mathbf{q}$), the operator $L_{\hat{\mathbf{Q}}}$ will leave its direction unchanged.

Proof:

$$\begin{aligned}\hat{\mathbf{Q}}\mathbf{v}\hat{\mathbf{Q}}^* &= \hat{\mathbf{Q}}(k\mathbf{q})\hat{\mathbf{Q}}^* \\&= (q_0^2 - \|\mathbf{q}\|^2) (k\mathbf{q}) + 2(\mathbf{q} \cdot k\mathbf{q})\mathbf{q} + 2q_0 (\mathbf{q} \times k\mathbf{q}) \\&= k (q_0^2 - \|\mathbf{q}\|^2) \mathbf{q} + 2k(\mathbf{q} \cdot \mathbf{q})\mathbf{q} + 2kq_0 (\mathbf{q} \times \mathbf{q}) \\&= k (q_0^2 - \|\mathbf{q}\|^2) \mathbf{q} + 2k(\|\mathbf{q}\|^2)\mathbf{q} + 0 \\&= [k (q_0^2 - \|\mathbf{q}\|^2) + 2k\|\mathbf{q}\|^2] \mathbf{q} \\&= [kq_0^2 - k\|\mathbf{q}\|^2 + 2k\|\mathbf{q}\|^2] \mathbf{q} \\&= [kq_0^2 + k\|\mathbf{q}\|^2] \mathbf{q} \\&= k (q_0^2 + \|\mathbf{q}\|^2) \mathbf{q} \\&= k\mathbf{q}\end{aligned}\tag{35}$$

Quaternions

Consider also that given that $L_{\hat{\mathbf{Q}}}$ is a linear operator, for any two vector $\mathbf{v}_1, \mathbf{v}_2 \in R^3$ and any $a_1, a_2 \in R$ we have that

$$L_{\hat{\mathbf{Q}}}(a_1\mathbf{v}_1 + a_2\mathbf{v}_2) = a_1L_{\hat{\mathbf{Q}}}(\mathbf{v}_1) + a_2L_{\hat{\mathbf{Q}}}(\mathbf{v}_2) \quad (36)$$

Quaternions

We can decompose a vector $\mathbf{v} \in \mathbb{R}^3$ as $\mathbf{v} = a + \mathbf{n}$, where a is the component along the vector $\mathbf{q}$ and $\mathbf{n}$ is the component normal to $\mathbf{q}$.

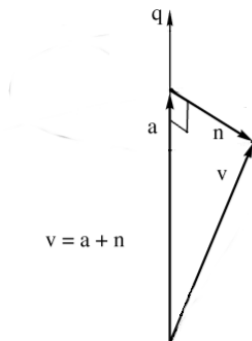


Figure 1: $\mathbf{v}$ can be decomposed as a vector a along $\mathbf{q}$ and the normal vector $\mathbf{n}$.

Quaternions

If we apply the operator $L_{\hat{\mathbf{Q}}}$ on $\mathbf{v}$, we have that:

$$\begin{aligned}L_{\hat{\mathbf{Q}}}(\mathbf{v}) &= L_{\hat{\mathbf{Q}}}(a + \mathbf{n}) \\&= L_{\hat{\mathbf{Q}}}(a) + L_{\hat{\mathbf{Q}}}(\mathbf{n}) \\&= a + L_{\hat{\mathbf{Q}}}(\mathbf{n}) \\&= a + (q_0^2 - \|\mathbf{q}\|^2) \mathbf{n} + 2(\mathbf{q} \cdot \mathbf{n})\mathbf{q} + 2q_0 (\mathbf{q} \times \mathbf{n}) \\&= a + (q_0^2 - \|\mathbf{q}\|^2) \mathbf{n} + 2q_0 (\mathbf{q} \times \mathbf{n}) \\&= a + (q_0^2 - \|\mathbf{q}\|^2) \mathbf{n} + 2q_0\|\mathbf{q}\| (\mathbf{u} \times \mathbf{n})\end{aligned}\tag{37}$$

Remember $\mathbf{u} = \frac{\mathbf{q}}{\|\mathbf{q}\|}$.

Quaternions

If we denote $\mathbf{n}_\perp = (\mathbf{u} \times \mathbf{n})$, equation (37) can be written as

$$L_{\hat{\mathbf{Q}}}(\mathbf{v}) = a + (q_0^2 - \|\mathbf{q}\|^2) \mathbf{n} + 2q_0 \|\mathbf{q}\| \mathbf{n}_\perp \quad (38)$$

From eq. (8) we know that $q_0 = \cos\left(\frac{\theta}{2}\right)$ and $\|\mathbf{q}\| = \sin\left(\frac{\theta}{2}\right)$,
hence

$$L_{\hat{\mathbf{Q}}}(\mathbf{v}) = a + \cos \theta \mathbf{n} + \sin \theta \mathbf{n}_\perp \quad (39)$$

Quaternions

If we denote $\mathbf{u}_\perp = (\mathbf{u} \times \mathbf{n})$, equation (37) can be written as

$$L_{\hat{\mathbf{Q}}}(\mathbf{v}) = a + (q_0^2 - \|\mathbf{q}\|^2) \mathbf{n} + 2q_0\|\mathbf{q}\|\mathbf{n}_\perp \quad (38)$$

From eq. (8) we know that $q_0 = \cos\left(\frac{\theta}{2}\right)$ and $\|\mathbf{q}\| = \sin\left(\frac{\theta}{2}\right)$,
hence

$$L_{\hat{\mathbf{Q}}}(\mathbf{v}) = a + \cos \theta \mathbf{n} + \sin \theta \mathbf{n}_\perp \quad (39)$$

Homework. Prove that eq. (38) and eq. (39) are equivalent.

Quaternions

Proof 1/2. If we consider:

$$q_0 = \cos\left(\frac{\theta}{2}\right), \quad q_0^2 = \cos^2\left(\frac{\theta}{2}\right), \quad ||\mathbf{q}|| = \sin\left(\frac{\theta}{2}\right) \quad \text{and}$$

$$\mathbf{q} = \mathbf{u} \sin\left(\frac{\theta}{2}\right)$$

Equation (38) can turn into this new equation:

$$\begin{aligned} L_{\hat{\mathbf{Q}}}(\mathbf{v}) &= a + \left[\cos^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right) \right] \mathbf{n} + \left[2\cos\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right) \right] \mathbf{n}_{\perp} \\ &= a + \cos \theta \mathbf{n} + \sin \theta \mathbf{n}_{\perp} \end{aligned}$$

Quaternions

Proof 2/2. Remember the following trigonometric identities:

$$\cos^2 \left(\frac{\theta}{2} \right) = 0.5 + 0.5 \cos \theta$$

$$\sin^2 \left(\frac{\theta}{2} \right) = 0.5 - 0.5 \cos \theta$$

Hence we have that

$$\begin{aligned} \cos^2 \left(\frac{\theta}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right) &= 0.5 + 0.5 \cos \theta - 0.5 + 0.5 \cos \theta \\ &= \cos \theta \end{aligned}$$

and we also use the following trigonometric identity:

$$2 \cos \left(\frac{\theta}{2} \right) \sin \left(\frac{\theta}{2} \right) = \sin \theta$$

Quaternions

We can prove that the terms $\cos\theta \mathbf{n} + \sin\theta \mathbf{n}_\perp$ can be seen as the components of a vector $\mathbf{m}$ in the frame formed by $\mathbf{n}$ and $\mathbf{n}_\perp$.

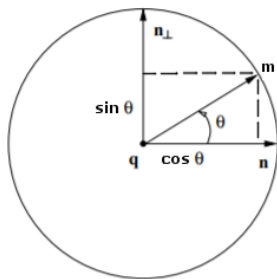


Figure 2: $\mathbf{m}$ results from rotating $\mathbf{v}$ about $\mathbf{q}$ through θ degrees.

Quaternions

Hence we can turn eq. (39) into this new equation:

$$\begin{aligned} L_{\hat{\mathbf{Q}}}(\mathbf{v}) &= a + \cos \theta \mathbf{n} + \sin \theta \mathbf{n}_{\perp} \\ &= a + \mathbf{m} \end{aligned} \tag{40}$$

$$= \mathbf{w} \tag{41}$$

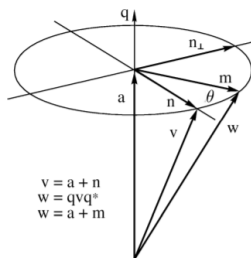


Figure 3: Rotation operator geometry.

Quaternions

From the previous analysis we can observe that after operating a vector $\mathbf{v} \in \mathbb{R}^3$ with the operator $L_{\hat{\mathbf{Q}}}$ we get a new vector $\mathbf{w} \in \mathbb{R}^3$ that results from the rotation of $\mathbf{v}$ through an angle θ about $\mathbf{q}$ as the axis of rotation.

This implies two more facts:

- we can use the operator $L_{\hat{\mathbf{Q}}}$ to rotate a vector through an angle θ about $\mathbf{q}$.
- a quaternion can encode the rotation of a vector or a coordinate frame similarly as a rotation matrix or the Euler angle representation does.

Quaternions

Let's find a *nicer* expression that allows us to rotate a vector through an angle θ about $\mathbf{q}$.

If we consider:

$$q_0 = \cos\left(\frac{\theta}{2}\right), \quad q_0^2 = \cos^2\left(\frac{\theta}{2}\right), \quad \|\mathbf{q}\| = \sin^2\left(\frac{\theta}{2}\right), \quad \mathbf{q} = \mathbf{u} \sin\left(\frac{\theta}{2}\right)$$

Equation (9) can turn into this new equation:

$$\begin{aligned} L_{\hat{\mathbf{Q}}}(\mathbf{v}) &= \left(\cos^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right) \right) \mathbf{v} \\ &\quad + 2 \left(\mathbf{u} \sin\left(\frac{\theta}{2}\right) \cdot \mathbf{v} \right) \mathbf{u} \sin\left(\frac{\theta}{2}\right) \\ &\quad + 2 \cos\left(\frac{\theta}{2}\right) \left(\mathbf{u} \sin\left(\frac{\theta}{2}\right) \times \mathbf{v} \right) \end{aligned} \tag{42}$$

Quaternions

Using the following trigonometric identities:

$$\cos^2 \left(\frac{\theta}{2} \right) = 0.5 + 0.5 \cos \theta$$

$$\sin^2 \left(\frac{\theta}{2} \right) = 0.5 - 0.5 \cos \theta$$

$$2 \sin \left(\frac{\theta}{2} \right) \cos \left(\frac{\theta}{2} \right) = \sin \theta$$

Equation (42) can be written as shown next:

$$L_{\hat{\mathbf{Q}}}(\mathbf{v}) = (\cos \theta) \mathbf{v} + (1 - \cos \theta) (\mathbf{u} \cdot \mathbf{v}) \mathbf{u} + \sin \theta (\mathbf{u} \times \mathbf{v}) \quad (43)$$

This equation can be used to find the coordinates of the vector w that results from rotating $\mathbf{v}$ through an angle θ about $\mathbf{u}$ as the axis of rotation.

$\mathbf{u}$ is a unit vector along $\mathbf{q}$.

Quaternions

Summary

A vector $\mathbf{v}$ can be rotated through an angle θ about $\mathbf{u}$ as the axis of rotation if we use the operator $L_{\hat{\mathbf{Q}}}(\mathbf{v})$, where

$$\begin{aligned} L_{\hat{\mathbf{Q}}}(\mathbf{v}) &= \hat{\mathbf{Q}}\mathbf{v}\hat{\mathbf{Q}}^* \\ &= (\cos \theta) \mathbf{v} + (1 - \cos \theta) (\mathbf{u} \cdot \mathbf{v}) \mathbf{u} + \sin \theta (\mathbf{u} \times \mathbf{v}) \end{aligned}$$

The quaternion $\hat{\mathbf{Q}} = q_0 + \mathbf{q} = \cos \left(\frac{\theta}{2} \right) + \mathbf{u} \sin \left(\frac{\theta}{2} \right)$ encodes a rotation through an angle θ about $\mathbf{q}$.

Quaternions

Example. Rotate the vector $\mathbf{v} = (1, 0, 0)$ through an angle $\theta = 90^\circ$ about the axis $\mathbf{u} = (0, 1, 0)$.

Quaternions

Example. Find the quaternion that defines the rotation of the vector $\mathbf{v}$ in the previous example.

Quaternions

Exercise. Rotate the vector $\mathbf{v} = (2, 0, 0)$ through an angle $\theta = 180^\circ$ about the axis $\mathbf{u} = (0, 1, 0)$.

Quaternions

Example. Find the quaternion that defines the rotation of the vector $\mathbf{v}$ in the previous example.

Quaternions

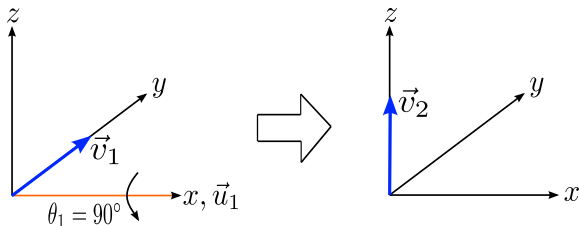
Exercise. Rotate the vector $\mathbf{v} = (1, 0, 0)$ through an angle $\theta = \frac{2\pi}{3}$ about the axis $\mathbf{u} = \frac{1}{\sqrt{3}}(1, 1, 1)$.

Quaternions

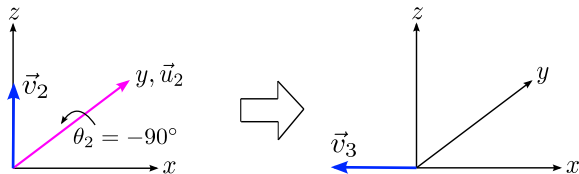
Example. Find the quaternion that defines the rotation of the vector $\mathbf{v}$ in the previous example.

Quaternions: composition of rotations

Suppose a vector $\vec{v}_1$ is rotated θ_1 degrees about axis $\vec{u}_1$:



and then it is rotated for a second time θ_2 degrees about axis $\vec{u}_2$:



Quaternions: composition of rotations

The two rotations can be composed by a single quaternion $\hat{\mathbf{Q}}_3$ as follows

$$\hat{\mathbf{Q}}_3 = \hat{\mathbf{Q}}_2 \otimes \hat{\mathbf{Q}}_1 \quad (44)$$

where

$\hat{\mathbf{Q}}_1$: Unit quaternion representing the first rotation

$\hat{\mathbf{Q}}_2$: Unit quaternion representing the second rotation.

It means that the orientation of $\vec{v}_3$ can be computed as seen before:

$$\vec{v}_3 = L_{\hat{\mathbf{Q}}_3}(\vec{v}_1) = \hat{\mathbf{Q}}_3 \mathbf{v}_1 \hat{\mathbf{Q}}_3^*$$

Quaternions: composition of rotations

Example. Get the new coordinates of vector $\vec{v} = [0, 1, 0]$ if

- it is rotated 270° about $\vec{u}_1 = [1, 0, 0]$,
- it is again rotated, but now 90° about $\vec{u}_2 = [0, 1, 0]$.

Quaternions: composition of rotations

Solution with method 1: Perform each rotation individually

$$\begin{aligned}\vec{m}_1 &= (\cos \theta_1) \mathbf{v} + (1 - \cos \theta_1) (\mathbf{u}_1 \cdot \mathbf{v}) \mathbf{u}_1 + \sin \theta (\mathbf{u}_1 \times \mathbf{v}) \\&= (\cos 270^\circ) [0, 1, 0] + (1 - \cos 270^\circ) ([1, 0, 0] \cdot [0, 1, 0]) [1, 0, 0] \\&\quad + \sin (270^\circ) ([1, 0, 0] \times [0, 1, 0]) \\&= (0)[0, 1, 0] + (1)(0)[1, 0, 0] + (-1)[0, 0, 1] \\&= [0, 0, 0] + [0, 0, 0] + [0, 0, -1] \\&= [0, 0, -1]\end{aligned}$$

Quaternions: composition of rotations

$$\begin{aligned}\vec{m}_2 &= (\cos \theta_2) \mathbf{m}_1 + (1 - \cos \theta_2) (\mathbf{u}_2 \cdot \mathbf{m}_1) \mathbf{u}_2 + \sin \theta_2 (\mathbf{u}_2 \times \mathbf{m}_1) \\&= (\cos 90^\circ) [0, 0, -1] + (1 - \cos 90^\circ) ([0, 1, 0] \cdot [0, 0, -1]) [0, 1, 0] \\&\quad + \sin (90^\circ) ([0, 1, 0] \times [0, 0, -1]) \\&= (0)[0, 0, -1] + (1)(0)[0, 1, 0] + (1)[-1, 0, 0] \\&= [0, 0, 0] + [0, 0, 0] + [-1, 0, 0] \\&= [-1, 0, 0]\end{aligned}$$

Quaternions: composition of rotations

Solution with method 2: Composing rotations

$$\begin{aligned}\hat{\mathbf{Q}}_1 &= \cos\left(\frac{270^\circ}{2}\right) + \sin\left(\frac{270^\circ}{2}\right) [i + 0j + 0k] \\ &= \cos(135^\circ) + \sin(135^\circ)i \\ &= -0.707 + 0.707i\end{aligned}$$

$$\begin{aligned}\hat{\mathbf{Q}}_2 &= \cos\left(\frac{90^\circ}{2}\right) + \sin\left(\frac{90^\circ}{2}\right) [0i + j + 0k] \\ &= \cos(45^\circ) + \sin(45^\circ)j \\ &= 0.707 + 0.707j\end{aligned}$$

Quaternions: composition of rotations

Composing both rotations:

$$\begin{aligned}\hat{\mathbf{Q}}_3 &= \hat{\mathbf{Q}}_2 \otimes \hat{\mathbf{Q}}_1 \\ &= [0.707 + 0.707i] \otimes [-0.707 + 0.707j] \\ &= -0.5 + 0.5i - 0.5j - 0.5k\end{aligned}$$

Compute the new coordinates of vector $\vec{v}$ using $\hat{\mathbf{Q}}_3$:

$$\begin{aligned}\vec{m}_3 &= \hat{\mathbf{Q}}_3[0, 0i + 1j + 0k]\hat{\mathbf{Q}}_3^* \\ &= [-0.5, 0.5i - 0.5j - 0.5k][0, 0i + 1j + 0k][-0.5, -0.5i + 0.5j + 0.5k] \\ &= [0.5, 0.5i - 0.5j - 0.5k][-0.5, -0.5i + 0.5j + 0.5k] \\ &= [0, -1, 0, 0] \rightarrow (\text{quaternion}) \\ &= [-1, 0, 0] \rightarrow (\text{vector})\end{aligned}$$

Appendix

Cross product between two vectors

The cross product $q \times p$ between two vectors q and p that belong to $\mathbb{R}^3$, results on a third vector c defined as follows

$$c = q \times p = \det \begin{bmatrix} i & j & k \\ q_1 & q_2 & q_3 \\ p_1 & p_2 & p_3 \end{bmatrix} \quad (45)$$

$$= (q_2 p_3 - q_3 p_2)i + (q_3 p_1 - q_1 p_3)j + (q_1 p_2 - q_2 p_1)k \quad (46)$$

The magnitud of the vector c can be computed as shown next

$$||c|| = ||q|| \ ||p|| \sin(\theta) \quad (47)$$

where θ is the angle between q and p .

Appendix

Algebraic properties of the dot product

Being A , B and C three vectors and c any real number:

Commutative property: $A \cdot B = B \cdot A$

Scalar multiplication property: $(cA) \cdot B = A \cdot (cB) = c(A \cdot B)$

Distributive property: $A \cdot (B + C) = A \cdot B + A \cdot C$

Appendix

Properties of the cross product

Being A , B and C three vectors and c any real number:

Anticommutativity: $A \times B = -B \times A$

Scalar multiplication: $(cA) \times B = c(A \times B) = A \times (cB)$

Distributivity: $A \times (B + C) = A \times B + A \times C$

The scalar triple product: $A \cdot (B \times C) = (A \times B) \cdot C$

The vector triple product: $A \times (B \times C) = (A \cdot C)B - (A \cdot B)C$